

3D & Spatial Visualizations

Presentation boards and illustrative views. Renders are artistic impressions; the analysis outputs are computed.

CONTENTS

- Visualisation Strategy and Image Provenance
- Board 1 concept
- Board 2 evidence
- Fig10 masterplan

VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository	github.com/wasim437/AI_SAFA
Live portal	wasim437.github.io/AI_SAFA/
Produced by	src/boards.py · src/plan.py · archive/withdrawn_visuals/README.md
Reproduce	<code>python run_analysis.py</code> · <code>python -m tests.test_pipeline</code>

How this document was produced

This submission is the output of a ten-phase process in which each phase is checked against the one before it. Nothing downstream is hand-drawn or hand-typed: change an input, re-run, and every chart, drawing and figure moves with it — or a test fails loudly. The phases behind this particular document are marked.

Phase 1 — Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

Phase 2 — Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

Phase 3 — Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

Phase 4 — Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

Phase 5 — Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m²

Phase 6 — Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

Phase 7 — Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

Phase 8 — User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

Phase 9 — AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

Phase 10 — Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

THE CODE AND DATA BEHIND THIS DOCUMENT

[src/boards.py](#)

[src/plan.py](#)

[archive/withdrawn_visuals/README.md](#)

REPRODUCE IT

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings          # section, elevation, planting
python -m src.boards            # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 38 checks
```

3D & Spatial Visualisations

What each image is, what produced it, and which ones are evidence rather than illustration

Three of this project's images were withdrawn for presenting invented data as measurement, and six photoreal renders were withdrawn for showing a park that is not the one in the drawings. What follows from that is a rule applied to every visual here: no image appears without stating what produced it and what it may be used to conclude.

Upload slot 05 — 3D & Spatial Visualisations

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m² · Al Safa 2, Dubai

Mohamed Wasim · Individual Applicant

Issued 03 August 2026

Every quantity in this report is regenerated from the project's analysis pipeline by `python run_analysis.py`. No figure quoted here is typed in by hand.

1. Three classes of image, never mixed

Class	What it is	What it may be used for
Technical drawing	Generated from the plan geometry in <code>src/plan.py</code> and drawn to scale	Measuring. Dimensions and areas taken off it are correct because they are the drawn polygon's own dimensions
Analysis output	Computed from project data by the pipeline — a chart of a model's behaviour or a simulation's result	Reading a result. It reports what the model found, at the accuracy stated on the figure
Artistic impression	AI-generated illustration of the design language	Judging character and atmosphere only. It is not evidence of anything, and no number may be taken from it

Every image in this submission carries its class on its own sheet. A juror should never have to guess which they are holding.

2. What is in this slot

Image	Class	Produced by
Board 1 — Concept	Presentation board	<code>src/boards.py</code> , reading <code>src/plan.py</code>
Board 2 — Evidence	Presentation board	<code>src/boards.py</code> , reading the trained-model metrics
Masterplan	Technical drawing	<code>src/plan.py</code> — areas are the shoelace area of each polygon

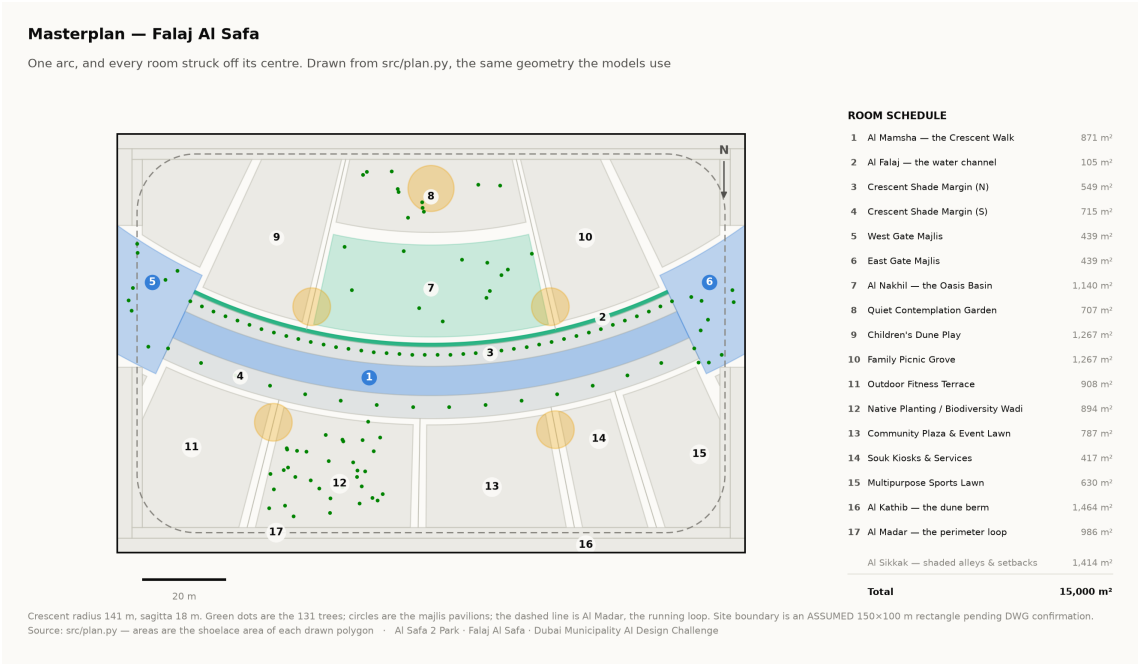
3. The photoreal renders, and why this slot does not currently contain any

Six photoreal renders were produced earlier in the project. Each was opened and compared against the masterplan, and all six failed: one showed a serpentine canopy over a large lagoon, one a dead-straight corridor — the superseded scheme the crescent replaced — one a vaulted pavilion planted with tropical species that contradict the five-species desert palette, and the remainder curved as an S rather than as one arc. All six were withdrawn.

The design they were supposed to show is specific and measurable: one continuous arc of 141 m radius bowing convex south, a 0.9 m water channel at its northern drip line, and 18 m of canopy over a 7 m walk. A render that shows a lake where the design has a rill is not a stylistic difference. It contradicts the sustainability argument, in which the entire water surface of the park is 105 m² — 0.70% of the site.

This slot is therefore presented with its measured drawings and its two boards, and without photoreal imagery, until renders exist that pass the acceptance test published with this project. A submission that shows a juror two different parks forfeits the authority of everything else in it. Board 1 marks the two views still in preparation rather than filling them with an image that would have to be captioned dishonestly.

4. The test every image must pass before it is admitted



The masterplan every visualisation in this submission must agree with. Technical drawing — to scale.

Every quantity above is regenerated by python run_analysis.py and asserted by python -m tests.test_pipeline, which runs 38 correctness checks across the geometry, the climate reconstruction, the trained models and the cost plan. Nothing in this report is typed in by hand.

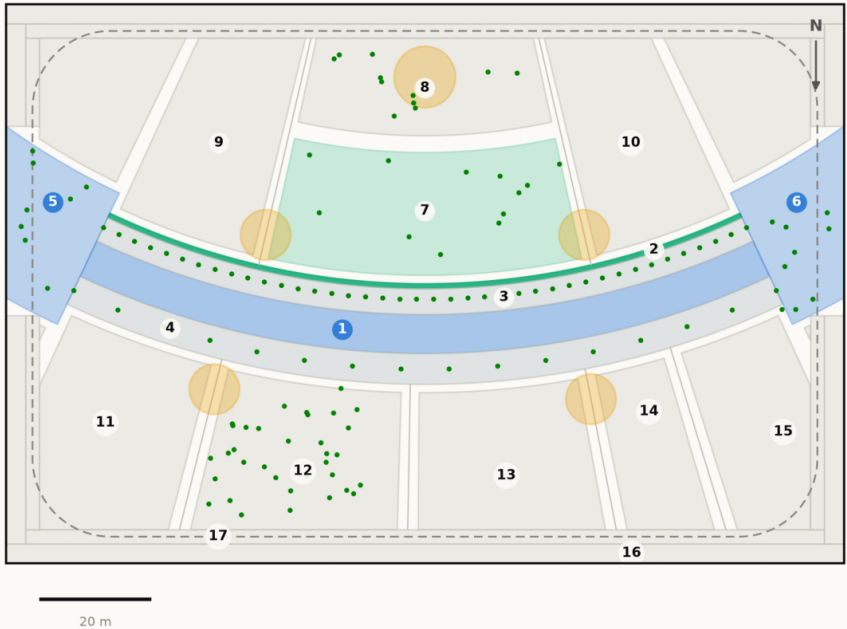
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 1 of 2 — CONCEPT

The concept, the plan, and the section that makes it work

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150x100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
Total		15,000 m²

The crescent from the north-west, at golden hour

Visualisation in preparation

Aerial view · AL_SAFa_MASTER_PROMPT.md prompt 01

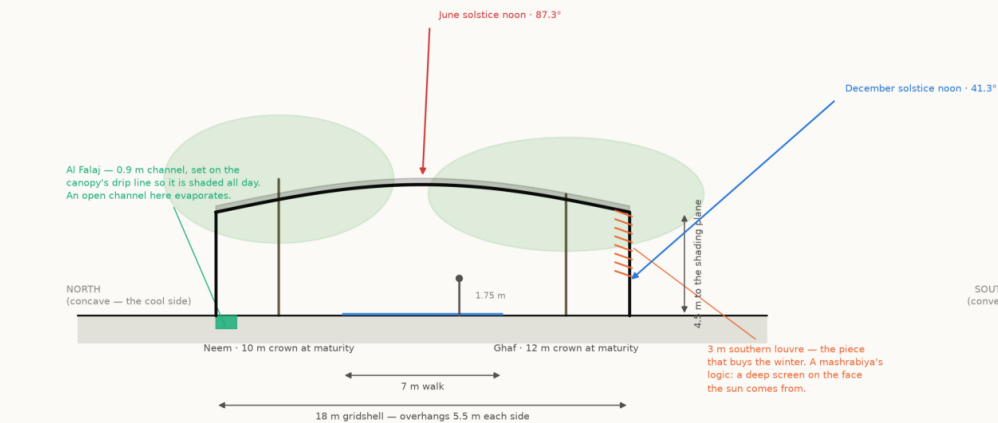
Beneath Al Hilal — the perforated soffit at eye level

Visualisation in preparation

Eye-level view · AL_SAFa_MASTER_PROMPT.md prompt 03

Section A-A — the Crescent Canopy at midspan

Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N

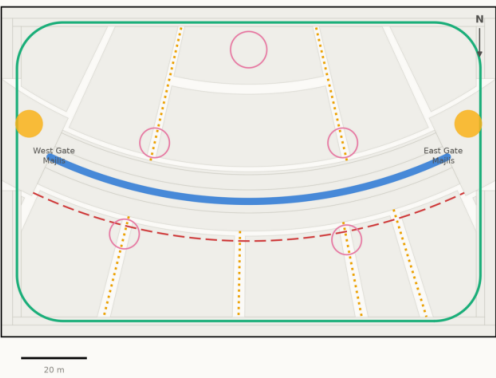


The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

One arc. Every room in the park is struck off its centre, so no room is a rectangle and every room faces the crescent square-on.
The bow is 18 m deep on a 141 m radius — swept against the 8,760-hour solar model, not drawn by eye. A straight canopy shades marginally more ground on average and leaves the walk with no shade at all for 330 hours a year; this one leaves it for 56.
Walk shaded 87.3% of daylight hours · 131 trees · 15,000 m² · every number reproducible with `python run\_analysis.py`

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



- Al Mamsha — primary route, shaded, step-free
- Al Sikkak — shaded alleys to the rooms
- Al Madar — 438 m running loop
- Service / emergency vehicle access
- Majlis — shaded rest, never more than 33 m from the walk

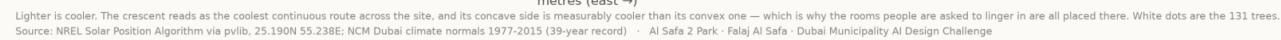
Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: srcplan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Presentation board.

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

Predicted summer comfort across the site

July afternoon heat index per square metre — the shade strategy, tested



What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



Planting plan – 131 trees

Drawn at mature canopy radius. The avenue is structural; it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.

Shade converts unusable daylight hours into usable ones

Share of the year's 4,402 daylight hours in each comfort band



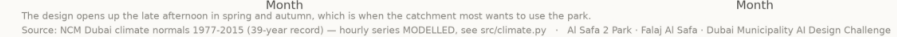
Annual shade coverage by zone type

Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



When is the park comfortable?

Mean heat index by hour and month — exposed today, shaded as designed



Elevation — the Crescent Canopy
60 m of the 124 m run, to scale, 6 m structural bay

60 m of the 124 m run, to scale, 6 m structural bay



Comfortable daylight hours rise from 44.5% to 64.6% — 20.1 points of the year handed back. Peak heat index falls 56.8 °C → 48.7 °C.

Site-wide mean shade is 34.1%, and that is stated as a position rather than hidden: this scheme makes a few places genuinely excellent instead of the whole site marginally better.

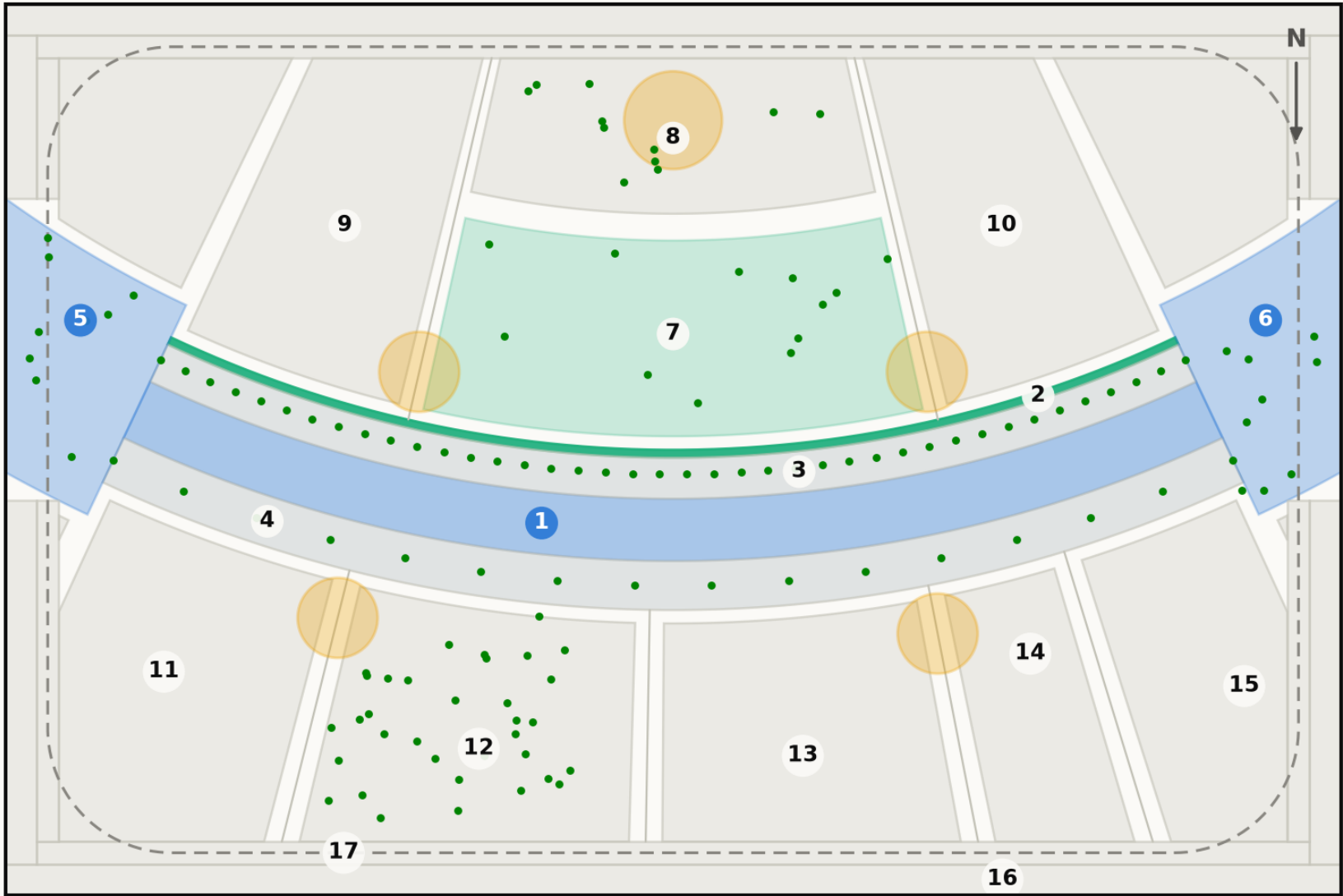
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette

Fig10 masterplan

Technical drawing — to scale. Geometry generated by src/plan.py; every area is the measured area of the drawn polygon.

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

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Al Sikkak — shaded alleys & setbacks		1,414 m²

Total 15,000 m²

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The complete project, and where to check it

Every one of the 129 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository github.com/wasim437/AI_SAFA
Live portal wasim437.github.io/AI_SAFA/

THE TWELVE UPLOAD FILES (12)

UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf	Design Narrative & Concept
UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf	Neighborhood Park Preliminary Design Masterplan
UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf	Concept Plans and Spatial Organization Diagrams
UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf	Key Sections & Elevations
UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf	3D & Spatial Visualizations
UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf	AI Methodology Report
UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf	User Experience & Activation Strategy
UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf	Sustainability Concept & Strategy
UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf	Material & Landscape Palette
UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf	Complete Design Report
UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf	Site Analysis & Human-Centric Research
UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf	One-minute Concept Animation

PROJECT DOCUMENTS (8)

AL_SAFA_MASTER_PROMPT.md	The whole design, stated for visualisation
DATA_SOURCES.md	Every source, its period, and its limitations
LINKS.md	Every public URL this submission points at
PROJECT_PLAN.md	Requirements, phases, status, and what is left
PUSH_TO_GITHUB.md	Project document
README.md	The design argument, written for a juror
RENDER_PROMPTS.md	Project document
EXPLAIN_THE_PROJECT/START_HERE.md	The project in plain language

THE WRITTEN REPORTS (11)

reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf	Generated from the live analysis
reports/pdf/Concept_Animation_Storyboard.pdf	Generated from the live analysis
reports/pdf/Phase2_Problem_Definition_Report.pdf	Generated from the live analysis
reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf	Generated from the live analysis
reports/pdf/Phase4_Concept_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase5_Masterplan_Development_Report.pdf	Generated from the live analysis
reports/pdf/Phase6_Detailed_Design_Report.pdf	Generated from the live analysis
reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf	Generated from the live analysis
reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf	Generated from the live analysis
reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf	Generated from the live analysis
reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf	Generated from the live analysis

THE ANALYSIS FIGURES (11)

figures/fig01_climate_and_comfort.png	Analysis output — computed from project data
figures/fig02_comfort_bands.png	Analysis output — computed from project data
figures/fig03_shade_by_zone.png	Analysis output — computed from project data
figures/fig04_site_comfort_map.png	Analysis output — computed from project data
figures/fig05_surrogate_performance.png	Analysis output — computed from project data
figures/fig06_feature_importance.png	Analysis output — computed from project data
figures/fig07_confusion_matrix.png	Analysis output — computed from project data
figures/fig08_microclimate_regimes.png	Analysis output — computed from project data
figures/fig09_diurnal_comfort.png	Analysis output — computed from project data
figures/fig10_masterplan.png	Analysis output — computed from project data
figures/fig11_cost_plan.png	Analysis output — computed from project data

The complete project — continued

THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation_crescent.png](#)
[design/visuals/elevation_crescent.png](#)
[design/visuals/facilities_crescent.png](#)
[design/visuals/planting_crescent.png](#)
[design/visuals/section_crescent.png](#)
[design/boards/board_1_concept.png](#)
[design/boards/board_2_evidence.png](#)

Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)
Technical drawing — generated from [src/plan.py](#)

THE ANALYSIS CODE (13)

[src/__init__.py](#)
[src/boards.py](#)
[src/climate.py](#)
[src/config.py](#)
[src/costing.py](#)
[src/dataset.py](#)
[src/drawings.py](#)
[src/figures.py](#)
[src/models.py](#)
[src/plan.py](#)
[src/solar.py](#)
[src/viz.py](#)
[run_analysis.py](#)

Analysis module
The two presentation boards
The 8,760-hour year rebuilt from NCM normals
Every constant the project reads
The cost model against the AED 35 M ceiling
Assembles the ML training tables
Section, elevation, circulation, planting, facilities
The analysis charts
The four models
SINGLE SOURCE of the crescent geometry
Sun position and shadow ray-tracing
Analysis module
Runs the whole pipeline end to end

THE BUILD AND SYNC TOOLS (11)

[tools/__init__.py](#)
[tools/build_docs.py](#)
[tools/build_links.py](#)
[tools/build_notebook.py](#)
[tools/build_reports.py](#)
[tools/build_submission_pdfs.py](#)
[tools/organize_repo.py](#)
[tools/report_content.py](#)
[tools/sync_film.py](#)
[tools/sync_portal.py](#)
[tools/sync_submission.py](#)

Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool
Build tool

THE DATA, AS ISSUED (7)

[data/raw/construction_unit_rates_aed.csv](#)
[data/raw/dubai_climate_normals_ncm.csv](#)
[data/raw/dubai_population_dsc_2023.csv](#)
[data/raw/site_zoning_schedule.csv](#)
[data/raw/sources.json](#)
[data/raw/species_water_carbon_rates.csv](#)
[data/raw/walk_catchment_rings.csv](#)

Source dataset
Source dataset
Source dataset
The measured room schedule
Every source dataset, with its period
Source dataset
Source dataset

THE DATA, AS PROCESSED (6)

[data/processed/cost_plan.csv](#)
[data/processed/hourly_climate_comfort_8760.csv](#)
[data/processed/masterplan_geometry.json](#)
[data/processed/planting_layout.csv](#)
[data/processed/schema.json](#)
[data/processed/spatial_grid_comfort.csv](#)

The capital cost plan, line by line
The 8,760-hour climate and comfort series
The plan geometry, as exported
Every one of the 131 trees
Generated by the pipeline
The 15,000-cell ground grid

The complete project — continued

THE TRAINED MODELS (3)

models/cost_summary.json	Model output
models/headline_metrics.json	The headline numbers
models/model_metrics.json	Trained-model metrics

THE TESTS (2)

tests/test_pipeline.py	38 correctness checks
tests/test_film.js	Every frame of the concept film

THE NOTEBOOK (1)

notebooks/AL_SAFI_2_PARK_COMPLETE_ANALYSIS.ipynb	The complete analysis, outputs embedded
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THE WEBSITE AND ANALYTICS PORTAL (9)

docs/index.html	The project website
docs/SUBMISSION_GUIDE.md	Published site
docs/_PORTAL/gallery_data.js	Published site
docs/_PORTAL/plan_geometry.js	Published site
docs/_PORTAL/portal.js	Published site
docs/_PORTAL/portal_data.js	Published site
docs/_PORTAL/selftest.js	Published site
docs/_PORTAL/DATA_AUDIT.md	Published site
docs/_PORTAL/README.md	Published site

THE COMPETITION BRIEF, AS ISSUED (7)

00_BRIEF/Ai Park - Master Plan (4).jpg	Dubai Municipality source document
00_BRIEF/Ai Park Scope of Work & General Terms & Conditions (5).pdf	Dubai Municipality source document
00_BRIEF/Ai Safa Park 2 Plan (5).dwg	Dubai Municipality source document
00_BRIEF/MAIN WEBSITE DETAILS.txt	Dubai Municipality source document
00_BRIEF/Neighborhood Parks Manual (4).pdf	Dubai Municipality source document
00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt	Dubai Municipality source document
00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt	Dubai Municipality source document

THE TWELVE SUBMISSION FOLDERS (12)

submission/01_Design_Narrative_Concept/MANIFEST.md	What is in the slot, and what produced each file
submission/02_Preliminary_Design_Masterplan/MANIFEST.md	What is in the slot, and what produced each file
submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md	What is in the slot, and what produced each file
submission/04_Key_Sections_Elevations/MANIFEST.md	What is in the slot, and what produced each file
submission/05_3D_Spatial_Visualizations/MANIFEST.md	What is in the slot, and what produced each file
submission/06_AI_Methodology_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/07_User_Experience_Activation_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/08_Sustainability_Concept_Strategy/MANIFEST.md	What is in the slot, and what produced each file
submission/09_Material_Landscape_Palette/MANIFEST.md	What is in the slot, and what produced each file
submission/10_Complete_Design_Report/MANIFEST.md	What is in the slot, and what produced each file
submission/11_Site_Analysis_Human_Centric_Research/MANIFEST.md	What is in the slot, and what produced each file
submission/12_Concept_Animation_Video/MANIFEST.md	What is in the slot, and what produced each file

WORKING HISTORY (9)

archive/withdrawn_visuals/README.md	Images withdrawn on purpose, and why
archive/phases/02_PHASE2_PROBLEM_DEFINITION/README.md	Working record
archive/phases/03_PHASE3_OPPORTUNITY_AND_OBJECTIVES/README.md	Working record
archive/phases/04_PHASE4_CONCEPT_DEVELOPMENT/README.md	Working record
archive/phases/05_PHASE5_MASTERPLAN_DEVELOPMENT/README.md	Working record
archive/phases/06_PHASE6_DETAILED_DESIGN/README.md	Working record
archive/phases/07_PHASE7_PERFORMANCE_AND_SUSTAINABILITY/README.md	Working record
archive/phases/08_PHASE8_USER_EXPERIENCE_AND_ACTIVATION/README.md	Working record
archive/phases/09_PHASE9_AI_WORKFLOW_AND_VISUALIZATION/README.md	Working record

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.